STUDY OF ADSORPTION OF METHYLENE BLUE DYE BY USING ACTIVATED CHARCOAL PREPARED FROM PLYWOOD

NAME:-RAKSHITA THAKRE

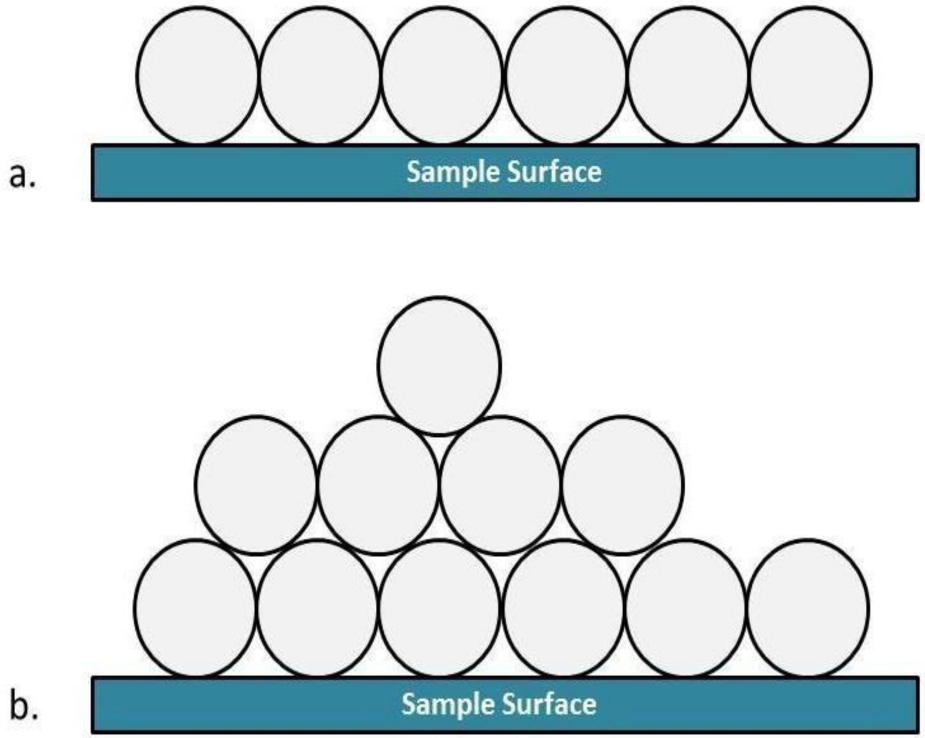
MSC Chemistry Sem-IV

Guided by-Dr.Rakshita Thakre

INTRODUCTION

- **Adsorption-** The term adsorption was first coined in 1881 by a German physicist named Heinrich Kayser.
- Adsorption is described as a surface phenomenon where particles are attached to the top layer of material.
- . It normally involves the molecules, atoms or even ions of a gas, liquid or solid in a dissolved state that is attached to the surface
- . Adsorption is a process that involves the accumulation of a substance in molecular species in higher concentration on the

surface. If we look at Hydrogen, Nitrogen and Oxygen, these gases adsorb on activated charcoal.


a.
Sample Surface
b.
Sample Surface

- Adsorption is mainly a consequence of surface energy.
- Generally, the surface particles which can be exposed partially tend to attract other particles to their site
- Adsorption is the densification of a fluid at an interface.
- The nature of the interface may be solid-liquid, gas liquid, liquid-liquid or solid-gas. In this work we focus on adsorption between a solid “adsorbent” and a gaseous “adsorptive species” which is denoted the “adsorbate” in the adsorbed phase

PLYWOOD CHARCOAL

•Charcoal generally made from plywood that has been burnt or charred while being deprived of oxygen

•Charcoal is a lightweight black carbon residue produce by heating plywood to remove all water and volatile constituent. Carbon rich product made by subjecting biomass substate to thermal destruction in the absence of air or limited amount of air process terminology.

• Charcoal is used in the manufacture of various object from crayons to filters, its most common use is as a fuel.

ACTIVATED CARBON

- **Activated carbon-**Activated Carbon is one of the most important types of adsorbents widely used as a tool in environmental protection in various industrial and domestic applications. It is a type of carbonaceous materials which can be used to adsorb materials.
- It normally differs from elemental carbon by the oxidation of carbon atoms that are present on the inner and outer surfaces. Currently, an important application of AC is the removal of polluted dyes from industrial wastewater.
- It is well known that industrial effluents produced from different industrial and human activities can cause high levels of environmental pollution as these wastes are highly coloured and contains large amounts of toxic organic species.
- The disposal of this coloured water into the surrounding environment results in the contamination by these polluted materials of the soil or water.

PHYSICAL ADSORPTION

- **Physical Adsorption**-This type of adsorption is also known as Physisorption. It is due to forces between adsorbate and adsorbent. For example, H_2 and N_2 gases adsorb on coconut charcoal

Characteristics of Physical Adsorption

1)This type of adsorption is caused by physical forces.

2) Physisorption is a weak phenomenon.

3) This adsorption is a multi-layered process.

4) Physical adsorption is not specific and takes place all over the adsorbent.

5) Surface area, temperature, pressure, nature of adsorbate effects physisorption.

CHEMICAL ADSORPTION

- **Chemical Adsorption-**This type of adsorption is also known as chemisorption. It is due to strong chemical forces of bonding type between adsorbate and adsorbent. We can take the example involving the formation of iron nitride on the surface when the iron heated in N_2 gas at 623 K.
- **Characteristics of Chemical Adsorption**
- 1)This type of adsorption is caused by chemical forces.
- 2)It is a very strong process.
- 3)This type of adsorption is almost a single -layered phenomenon.

- 4)Chemisorption is highly specific and takes place at reaction centers of adsorbent.
- 5)Surface area, temperature, nature of adsorbate effects chemisorption.

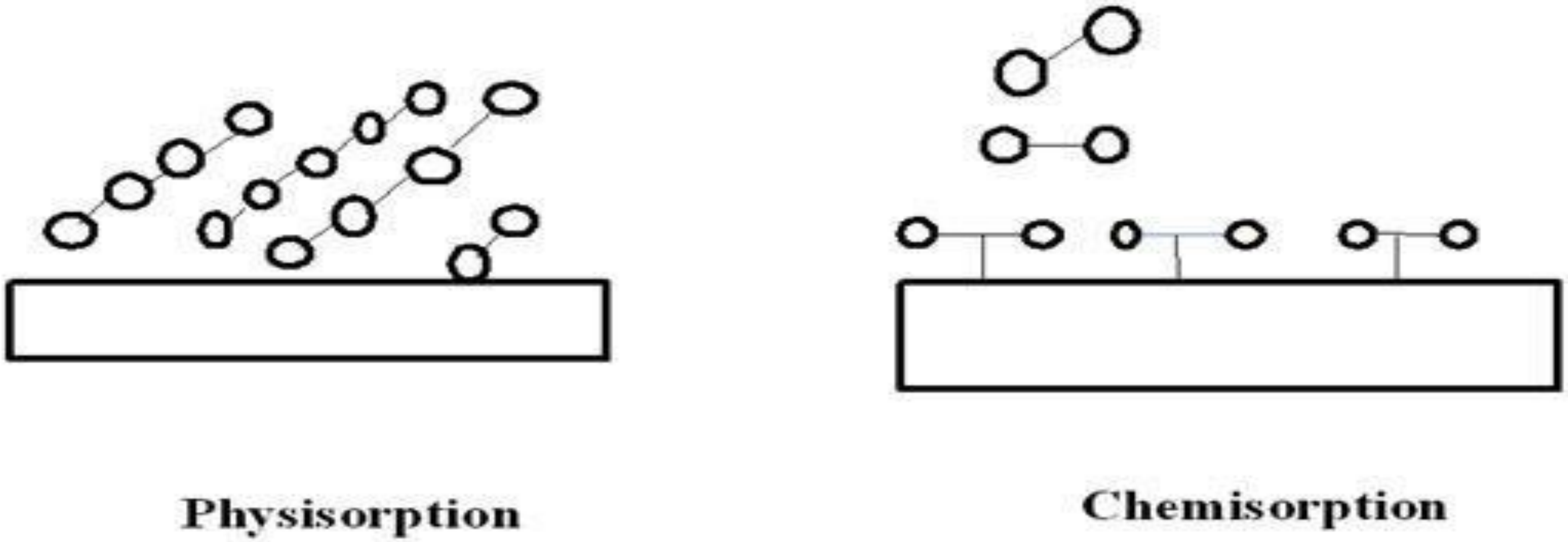


Figure 1: Mechanism of Physisorption & Chemisorption

FREUNDLICH ADSOPTION ISOTHERM

•Freundlich proposed this relation to show a relation between the extent of adsorption and pressure. Freundlich Adsorption Isotherm obeyed when the adsorbate forms a mono molecular layer on the surface of the adsorbent.

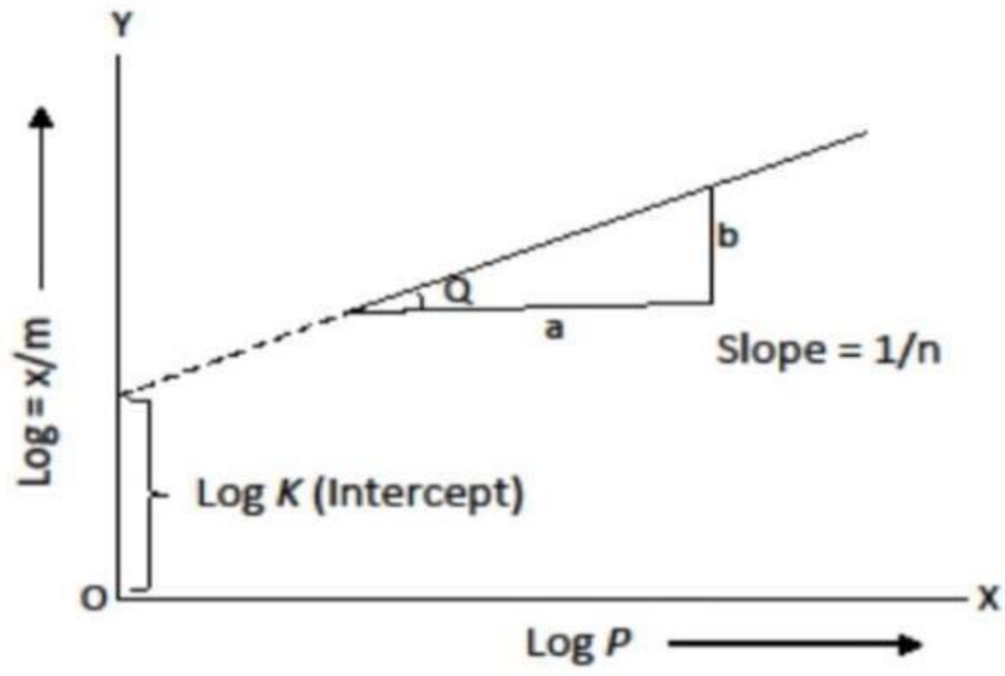
$$\mathbf{X/m=k(p)^{1/n}}$$

•log x/m=log k+(1/n)log(p)

•Where,

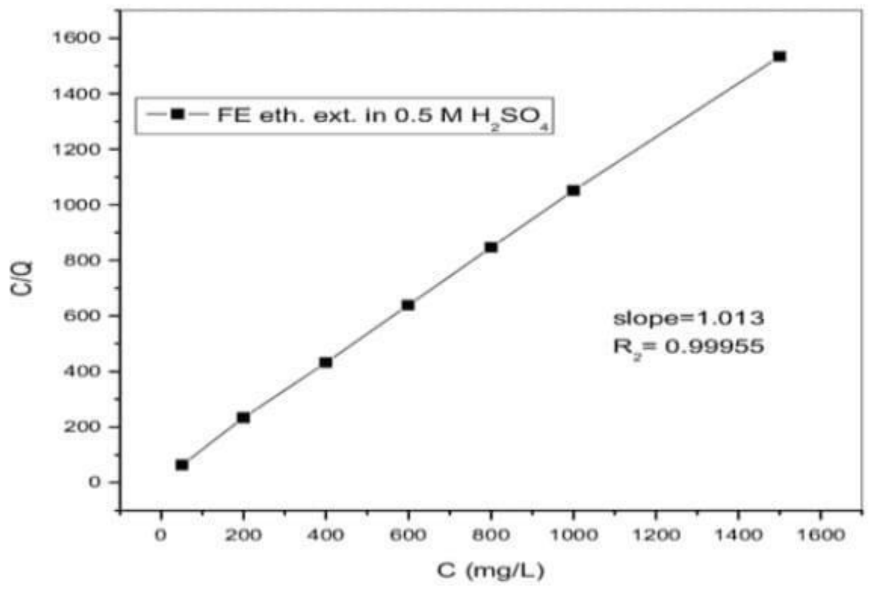
•X:the amount of adsorbate m:mass of adsorbent p:pressure •C:concentration of adsorbate

• K and n:constant,n>1 always



LANGMUIR THEORY

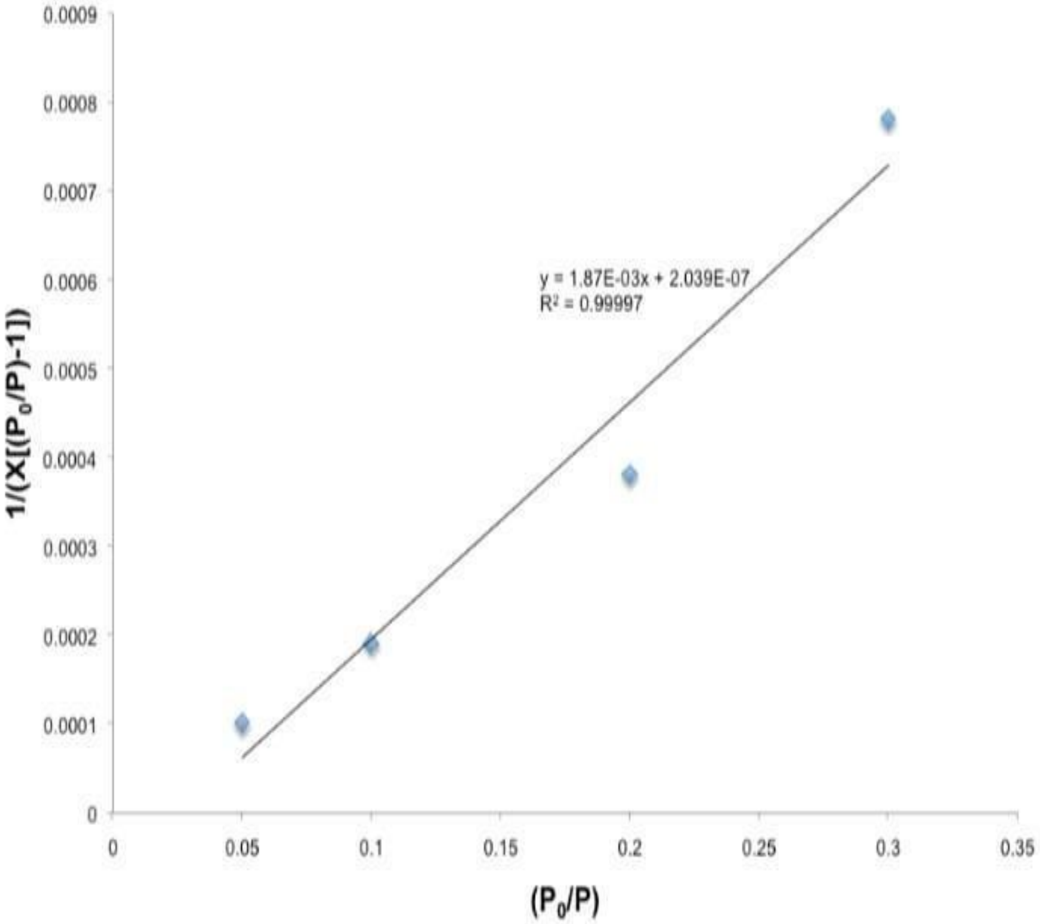
- In 1916,Irving Langmuir presented his model for the adsorption of species into simple surfaces. He hypothesized that a given adsorbent sites to which an adsorbate can get adsorbed, either by physisorption or chemisorption. These sites are called elementary sites. Langmuir theory could only predict linear adsorption at low adsorption densities and at maximum surface coverage at higher solute metal concentration. The Langmuir adsorption isotherm has the form



- **θ=KP/KP+1**
- Where,
- θ:the fraction of the surface covered by the adsorbed molecule.
- K:an equilibrium constant that is known as adsorption coefficient p:the pressure

BET ADSORPTION ISOTHERM

- Proposed by Brunauer, Emmett and Teller in 1938,this theory works for both partial and multilayer adsorption.
- It aims to explain the physical adsorption of gas molecules on a solid surface.
- Surface areas are commonly termed as BET surface areas are obtained by applying the theory of Brunauer, Emmett and Teller20 to nitrogen adsorption isotherms measured at 77K.
- The solid a has uniform BET surface areas and adsorption at one area does not affect adsorption at neighbouring areas.



FORMULA OF BET ADSORPTION ISOTHERM

$$\frac{P}{V(P^o - p)} = \frac{1}{V_m C} + \frac{c-1}{V_m C}\left(\frac{P}{P^o}\right)$$

where, V = volume of gas adsorbed at pressure 'p' and temperature 'T'

P^o = saturated vapor pressure

V_m = volume of gas for monolayer formation

$C = e\left(\frac{E_1 - E_L}{RT}\right)$ is a constant at temperature 'T'.

Here, E_1 = Heat of adsorption of gas in the formation of monolayer

E_L = Heat of liquefaction of gas

COLORIMETRY METHOD

- In physical and analytical chemistry, colorimetry or colourimetry is a technique used to determine the concentration of coloured compounds in solution.
- A colorimeter is a device used to test the concentration of a solution by measuring its absorbance of a specific wavelength of light
- To use the colorimeter, different solutions must be made, including a control or reference of known concentration. The color or wavelength of the filter chosen for the colorimeter is extremely important, as the wavelength of light that is transmitted by the colorimeter has to be the same as that absorbed by the substance being measured.

PREPARATION OF ACTIVATED CHARCOAL FROM PLYWOOD

1) Take a raw material of plywood wrap it with under Aluminium foil.

2) Put it in muffle furnace at 600°C temperature for 2 hour

3) Chemical activation with potassium hydroxide (KOH) was done,Then obtained material is mixed with potassium hydroxide.(1:1)

4) The activation was formed in a muffle furnace at 600°c for 1 hours.

5) Product obtained was watched with 10% HCL and after that deionization with water until neutral Ph.

6) At last the sample were dried at 110 °C within 12 hours


MUFFLE FURNACE

•Preparation of charcoal:-Accurately weighed samples of dried Plywood (weighed with analytical balance, Sartorius Basic) were carbonized in a closed crucible (size 105/73 and 102/70) at 500℃ in furnace (Fisher Scientific Isotemp Muffle Furnace) for 1 h.The charcoal product was ground and sieved to 2 mm size. For each system 0.10,0.12,0.14,0.16,0.18,0.20 g of activated charcoal powder was taken.

•Adsorption of Methylene Blue:-Activated carbon is amphoteric material, which can be positively or negatively charged depending on the solution pH. Attraction between activated carbon and anionic or cationic guest materials is mainly related to the surface characteristics[4].Methylene blue (C16H18N3SCl) is a cationic dye with the estimated dimensions of 1.43 nm × 0.61 nm × 0.4 nm and its adsorption by activated carbon is very susceptible to solution pH.[5].The methylene blue adsorption is lower than that of activated carbons prepared at 500℃and 700℃.[4].

METHYLENE BLUE DYE

- Methylene blue is a colourful organic chloride salt compound used in medicine. Methylene blue is a redox dye with clinical uses applicable to operating treasure and intensive care setting it uses include tissues staining and in the treatment of cyanide poisoning . methylene blue dye is a commonly used to exhibit antioxidant, anti depression, and the protective properties .
- Methylene blue is the cartionic stain and it is used to stain in the nucleus because it has a positive ionic charge it Will interact with tissue cell nuclei that have negative charge

OBSERVATION AND CALCULATION

Wavelength (m)	Absorbance
450	0.34
470	0.42
510	0.63
520	1.02
540	1.03
570	1.17
600	1.12
670	0.39

Table3.1 Selection of wavelength of maximum absorbance λmax = 570 m(absorbance=1.17)

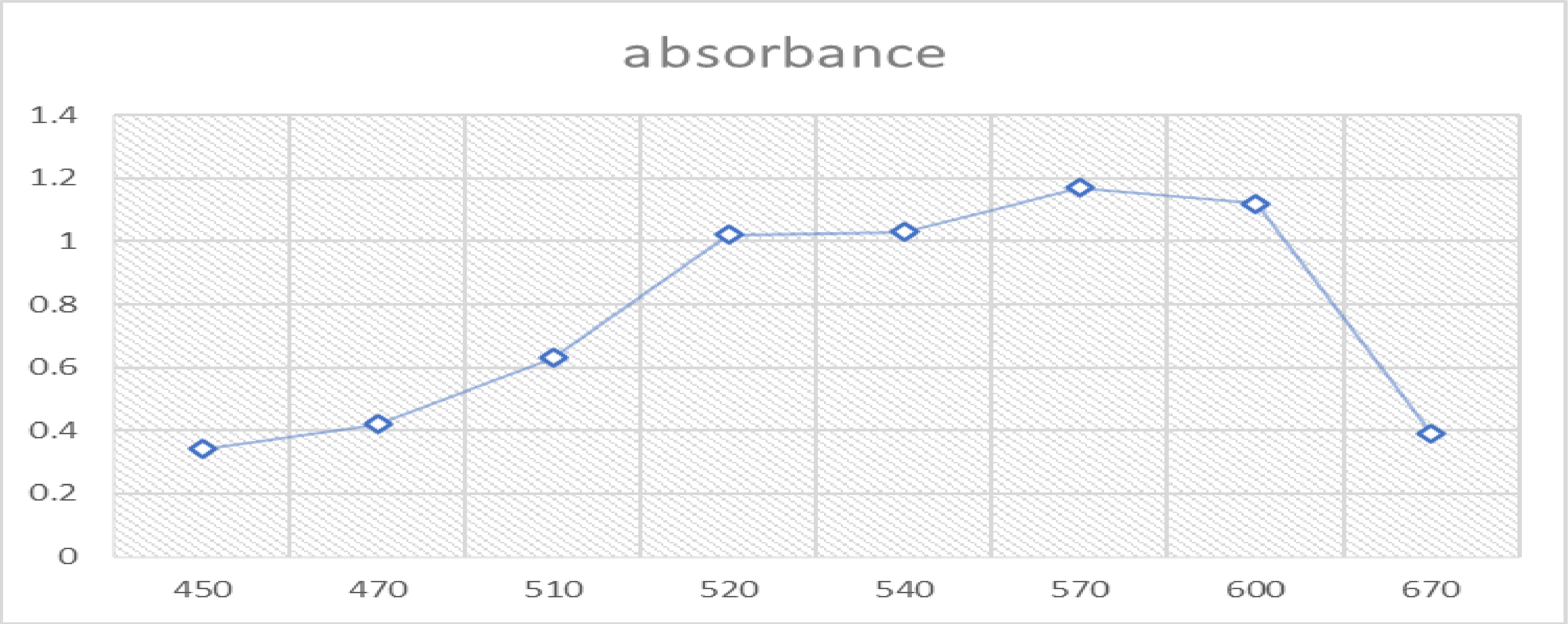


Figure 3.2 Absorbance of methylene blue

Concentration of MB dye%	Wavelength	Absorbance
0.10	570	1.50

0.12	570	1
0.14	570	1
0.16	570	1
0.18	570	1
0.20	570	1

Table 3.3 Adsorption on Plywood based activated carbon as function of methylene blue

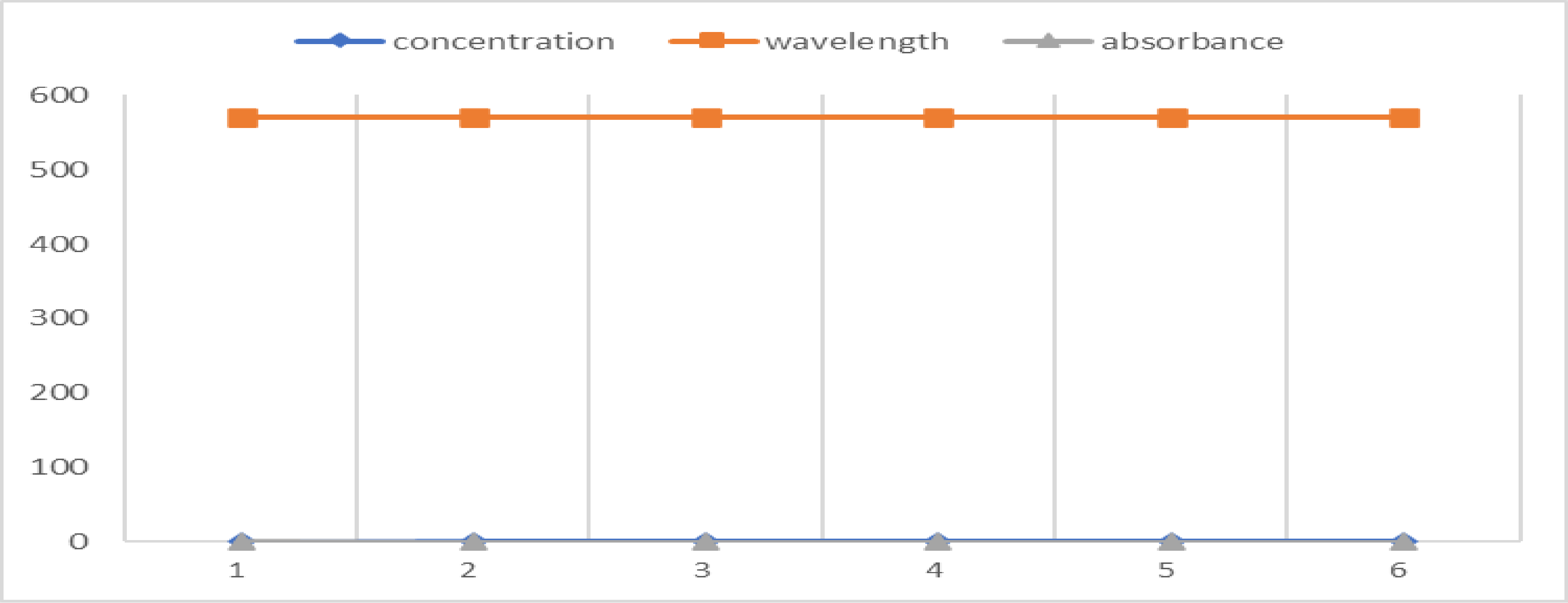

concentration
wavelength
absorbance
600
500
400
300
200
100
0
1
2
3
4
5
6

Figure 3.4

CONCLUSION AND RESULT

- From table no. 3.3 change in absorbance because of change in concentration of methylene blue dye could be noted.
- Proximate Analysis of Plywood Charcoal and percent yield of activated carbon is 2.18%.ash which makes it suitable precursor for obtaining activated carbon.The percent yield of Plywood based activated carbon decreases as activation temperature increases from 500^0C and above

REFERENCES

- American Standards for Testing of Materials, “Standard test method for volatile matter content of activate arbon,” ASTM D 5832-95, American Standards for Testing of Materials, 1996
- American Standard of Testing Material, “Standard test method for fixed carbon in activate carbon,”ASTMD3172-89,American Standard of Testing Material, 1994.
- Hindawi Publishing Corporation the Scientific World Journal Volume 2015, Article ID 415961
-]https://www.researchgate.net/publication/253853317_Calorimeter_for_adsorption _energies_of_larger_molecules_on_single_crystal_surfaces • . www.wikipedia.com
- www.google.com

THANK YOU